OPERATING SYSTEMS

ASSIGNMENT 9 | SOLUTION REPORT

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Course: Operating Systems | Topics: virtual memory and page replacement

1 | Printing a process page table

The vmprint system call walks the calling process's two-level page table and reports each present user mapping below KERNBASE. Every entry includes its virtual address, physical frame address, and present, writable, and user flags (P, W, and U). The routine also counts mapped user pages and allocated user page-table pages. pttest prints the initial layout, aligns the process break, grows it by four pages, touches one byte in each new page, and prints the layout again. Touching the pages is necessary after lazy allocation is enabled. The final count should increase by exactly four user pages.

2 | Lazy allocation and page-fault handling

Positive sbrk requests enlarge the process's recorded size without immediately reserving physical memory. When a user instruction first accesses an address inside that range, the page-fault handler rounds the address down to its page boundary, allocates a zero-filled frame, maps it with user and write permissions, increments the process fault counter, and returns so the instruction can retry. Invalid addresses and allocation failures mark the faulting process for termination. deallocuvm and copyuvm must skip absent pages safely. faulttest reserves 32 pages, writes to five distinct pages, and reads the count through getfaults; the expected result is five. A child then makes an out-of-range access so the parent can confirm that only the child is terminated.

3 | FIFO and LRU page replacement

The pagerepl program accepts a frame count and a reference string, then simulates both policies. FIFO replaces the frame resident longest, using a circular pointer. LRU replaces the frame whose most recent reference is oldest. For each reference, the program displays frame contents and marks a hit or fault. It reports fault totals and hit ratios for 3, 4, and 5 frames. For the required 20-reference string, the three-frame run should produce 15 FIFO faults and 12 LRU faults.

EXPECTED RESULTS FOR THE REQUIRED 20-REFERENCE STRING

Frames	FIFO faults / hit ratio	LRU faults / hit ratio
3	15 / 25%	12 / 40%
4	10 / 50%	8 / 60%
5	9 / 55%	7 / 65%

Implementation and interpretation

The kernel work adds syscall plumbing and a per-process page-fault counter, while the user programs exercise the requested behavior. Page-table output makes virtual-to-physical mappings visible; the lazy-allocation count shows that physical pages are created on first use; and the replacement simulation shows how policy and frame count affect faults. The hit ratios are calculated over all 20 references. These simulation results are separate from the lazy-allocation page-fault count.

Submission note: Integrate the kernel changes into the target xv6 x86 tree and capture actual terminal screenshots for all three programs. The accompanying transcript is illustrative, not a screenshot from a running xv6 kernel.